

SUMMARY

This video explains the theory of advanced relation types, closure operations, and the deep structural connection between equivalence relations and partitions.

Core themes include:

Logical counting of constrained relations.

Formal definitions of complement, inverse, and composition.

Construction of reflexive, symmetric, and transitive closures.

One-to-one correspondence between equivalence relations and partitions.

I. COUNTING SPECIFIC TYPES OF RELATIONS

Let A be a set with $|A| = n$.

Total ordered pairs in $A \times A = n^2$.

We analyze diagonal pairs (a, a) and non-diagonal pairs (a, b) , where $a \neq b$.

1.1 Symmetric Relations

Constraint

If $(a, b) \in R$ then $(b, a) \in R$.

Diagonal Pairs

Each diagonal pair may be present or absent.

Choices = 2^n .

Non-Diagonal Pairs

For each unordered pair $\{a, b\}$, there are 2 possibilities:

Both (a, b) and (b, a) present

Neither present

Number of such unordered pairs = $n(n - 1)/2$.

Total symmetric relations:

$$2^n \times 2^{n(n - 1)/2}$$

$$= 2^{n(n + 1)/2}$$

1.2 Antisymmetric Relations

Constraint

If (a, b) and (b, a) both belong to R , then $a = b$.

Diagonal Pairs

No restriction.

Choices = 2^n .

Non-Diagonal Pairs

For each unordered pair $\{a, b\}$, three valid options:

Only (a, b) present

Only (b, a) present

Neither present

Both present is forbidden.

Total antisymmetric relations:

$$2^n \times 3^{(n(n-1)/2)}$$

1.3 Asymmetric Relations

Constraint

If $(a, b) \in R$ then $(b, a) \notin R$.

Diagonal Pairs

Not allowed.

Only 1 choice.

Non-Diagonal Pairs

Same 3 options per unordered pair.

Total asymmetric relations:

$$3^{(n(n-1)/2)}$$

II. RELATION OPERATIONS

2.1 Complement of a Relation

Let $R \subseteq A \times B$.

Complement:

$$R^c = (A \times B) - R$$

2.2 Inverse of a Relation

If $R \subseteq A \times B$:

$$R^{-1} = \{(b, a) \mid (a, b) \in R\}$$

Defined for all relations.

2.3 Composition of Relations

Let:

$$R: X \rightarrow Y$$

$$S: Y \rightarrow Z$$

Composite relation:

$$(a, c) \in R \circ S$$

If there exists $b \in Y$ such that:

$$(a, b) \in R \text{ and } (b, c) \in S$$

III. RELATION CLOSURES

A closure is the smallest relation containing R that satisfies a given property.

Closure Type	Construction
Reflexive	$R \cup \Delta A$
Symmetric	$R \cup R^{-1}$
Transitive	Iteratively add required transitive pairs

$$\Delta A = \{(a, a) \mid a \in A\}$$

3.1 Transitive Closure Procedure

Step 1

Identify pairs (a, b) and (b, c) .

Step 2

Add (a, c) if missing.

Step 3

Repeat until no new pairs are generated.

Important

Newly added pairs may create further transitivity requirements.

IV. EQUIVALENCE RELATIONS

A relation is an equivalence relation if it is:

Reflexive

Symmetric

Transitive

4.1 Examples

Universal Relation

$A \times A$

Largest equivalence relation.

Diagonal Relation

ΔA

Smallest equivalence relation.

Blood Group Relation

"Has the same blood group as"

Reflexive

Symmetric

Transitive

Brother Of Relation

Fails reflexivity.

Fails symmetry.

Fails transitivity.

Hence not an equivalence relation.

V. EQUIVALENCE CLASSES

For $x \in A$:

$$[x] = \{y \in A \mid (x, y) \in R\}$$

Properties

Elements in same class are mutually related.

Two equivalence classes are either identical or disjoint.

VI. PARTITIONS OF A SET

A partition of A is a collection:

$$\{A_1, A_2, \dots, A_k\}$$

Such that:

Condition	Requirement
Disjoint	$A_i \cap A_j = \emptyset$ for $i \neq j$
Exhaustive	$A_1 \cup A_2 \cup \dots \cup A_k = A$
Non-Empty	Each $A_i \neq \emptyset$

VII. CORRESPONDENCE BETWEEN EQUIVALENCE RELATIONS AND PARTITIONS

From Relation to Partition

Distinct equivalence classes form a partition of A .

From Partition to Relation

Given partition $\{A_i\}$:

$$R = \bigcup (A_i \times A_i)$$

Conclusion

Number of equivalence relations on A

= Number of partitions of A

This establishes a one-to-one correspondence between equivalence relations and partitions.